



SAP INTEGRATION SUITE · INTEGRATION PACKAGE

SAP S/4HANA Integration with Keboola

Integration Guide

Setting up, running and troubleshooting the three integration flows, the value mapping and the Keboola adapter on SAP Integration Suite.

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Overview

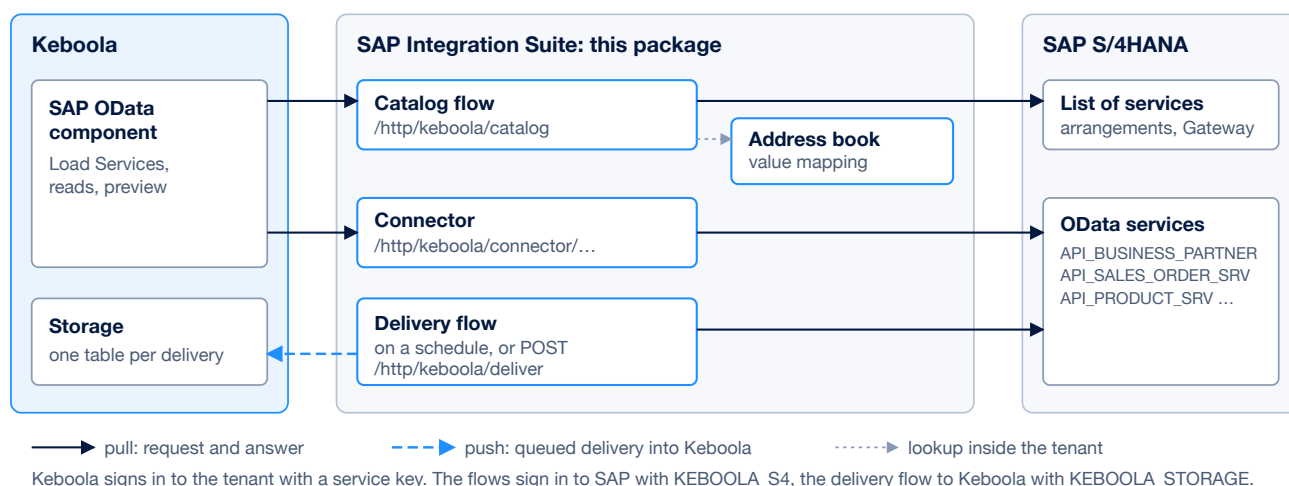
This package brings data from SAP S/4HANA into Keboola. It runs on your SAP Integration Suite tenant and works in two ways:

- **Pull.** Keboola asks, SAP answers. Keboola's SAP OData component reads through the package whenever a Keboola job runs.
- **Push.** Your tenant reads SAP on a schedule and writes the rows into a Keboola table itself. Nothing in your tenant has to be reachable from Keboola, and a run waits in a queue when Keboola is briefly unavailable.

You can use either, or both. Most installations explore and develop with pull, and use push for the tables that matter in production. Everything that differs from one installation to the next is a field on the **Configure** screen of a flow. You never need to open the flow editor. The one artifact you may edit is the address book (§3).

What is in the package

Artifact	Called in this guide	What it does	Do you need it?
Query Business Data from SAP S4HANA to Keboola	the connector	Passes each read from Keboola to SAP, and SAP's answer back	Required for pull
Query Available Services from SAP S4HANA	the catalog flow	Lists the OData services of your SAP system. It fills <i>Load Services</i> , the service picker in Keboola	Optional
SAP S4HANA Service Addresses for Keboola	the address book	A value mapping with the paths and titles of SAP's released OData services. The catalog flow reads it	Optional. Deploy it with the catalog flow on S/4HANA Cloud
Deliver Business Data from SAP S4HANA to Keboola	the delivery flow	Reads SAP on a schedule and writes into one Keboola table	Required for push. Needs JMS message queues
Keboola adapter	the adapter	A receiver adapter for integration flows you build yourself (§10)	Optional. None of the flows uses it



1. Set up

What you need

1. **An SAP Integration Suite tenant** with Cloud Integration. The delivery flow also needs JMS message queues. On an edition without them you can still use pull, but do not deploy the delivery flow.
2. **A user on SAP S/4HANA that may read the APIs you want.** On S/4HANA Cloud that is a communication user with a communication arrangement per scenario (§2), plus `SAP_COM_0A07` for the catalog flow.
3. **A Keboola project.** For the delivery flow also a bucket for the table, and a Storage token with write access to that bucket.

The steps

Two sign-ins are set up: Keboola signs in to your tenant with a service key (step 2), and the flows sign in to SAP with a security material (step 3).

1. **Copy the package into your tenant** from the SAP Business Accelerator Hub (in the tenant, under **Discover**).
2. **Create the service key Keboola signs in with.** In the BTP cockpit, create a service instance of *SAP Process Integration Runtime*, plan `integration-flow`, with the parameters `{"roles":["ESBMessaging.send"]}` (or a role of your own, below). Create a service key on it. Keboola needs three values of the key: `url`, `clientid` and `clientsecret`. The `url` is your tenant's runtime address, written `https://<runtime host>` in this guide.
3. **Create the security material the flows sign in to SAP with.** Under **Monitor** → **Manage Security** → **Security Material**, create an entry named `KEB00LA_S4`. Its type follows the sign-in method (§2). Appendix A has the clicks.
4. **Push only: create the security material for Keboola.** *User Credentials* named `KEB00LA_STORAGE`, with any user name and the Storage token as the password.
5. **Deploy the address book** if you use the catalog flow on S/4HANA Cloud. It has nothing to configure.
6. **Configure each flow you use, then deploy it.** Under **Design**, open the package and choose **Configure** on the flow. Every flow needs `S4_HOSTNAME` (§2). The delivery flow also needs `KEB00LA_STACK_URL`, `KEB00LA_TABLE_ID` and what to read (§6).
7. **Pull: set up the component in Keboola (§5).** It works when *Preview Data* shows rows.
8. **Push: start one run by hand (§6).** It works when the run shows `DELIVERED` and the table is in Keboola.

A flow tells you when it is not configured. Its first call answers `500 CONFIG_ERROR` and names the setting, for example `S4_HOSTNAME` is not configured on this integration flow.

Let only Keboola in: use a role of your own

The flows ship asking for the role `ESBMessaging.send`, which SAP predefines in every tenant. Every caller that holds it can call every flow that asks for it. With a role of your own, only the callers you give it to get in. SAP's design guidelines recommend it:

1. **Monitor** → **Manage Security** → **User Roles** → **Add**. Name the role, for example `KeboolaIntegration.Send`.
2. Create the service instance of step 2 with that role instead, and give Keboola the key.
3. Set `SENDER_ROLE` to your role on the Configure screen of each flow, and deploy.

A caller without the role is answered `403` before the flow runs, and no run is written. A typo in `SENDER_ROLE` is not caught at deploy. It shows only as that `403`.

2. Connect to SAP S/4HANA

All three flows have the same connection settings.

Parameter	Shipped	Meaning
<code>S4_HOSTNAME</code>	empty	Address of the SAP system with its protocol, for example <code>https://my123456-api.s4hana.cloud.sap</code> . Must start with <code>https://</code> , except behind an SAP Cloud Connector (below)
<code>S4_AUTH_METHOD</code>	<code>OAuth2 Client Credentials</code>	How the flow signs in: <code>OAuth2 Client Credentials</code> , <code>Basic</code> or <code>Client Certificate</code> . Chosen from a list
<code>S4_CREDENTIAL_ALIAS</code>	<code>KEB00LA_S4</code>	Name of the security material with the sign-in details. Not used with a client certificate
<code>S4_PRIVATE_KEY_ALIAS</code>	empty	Alias of the key pair in the keystore. For <code>Client Certificate</code> only
<code>S4_PROXY_TYPE</code>	default	default for a system on the internet, <code>sapcc</code> for one behind an SAP Cloud Connector
<code>S4_CC_LOCATION_ID</code>	empty	Cloud Connector location. Only if you run more than one
<code>S4_SAP_CLIENT</code>	empty	Client number of a multi-client on-premise system, sent as <code>sap-client</code> with every call. Leave empty on S/4HANA Cloud
<code>HTTP_TIMEOUT_MS</code>	50000 connector, 10000 catalog flow, 120000 delivery flow	How long one call may take, in milliseconds
<code>SENDER_ROLE</code>	<code>ESBMessaging.send</code>	The role a caller must hold (§1)
<code>AIR_KEY</code> , <code>AIR_HEADER_NAME</code>	empty	SAP's integration key and its header name. Leave empty unless Keboola gives you values (§8)

Without a Cloud Connector, the flows refuse an `http://` address with `500 CONFIG_ERROR`: your SAP credential and every row of data would cross the network in clear text.

Which sign-in method

To change the method, choose it in `S4_AUTH_METHOD`, name the matching security material, and deploy. Appendix A has the steps for each method.

- **Behind an SAP Cloud Connector: Basic only.** SAP's HTTP adapter allows no client certificate on that route, and a token request to an on-premise address fails there. The credential stays inside the encrypted tunnel.
- **Everywhere else: any of the three.** OAuth 2.0 Client Credentials is the shipped method. On S/4HANA Cloud only some communication scenarios offer it. On-premise, the ABAP OAuth server issues the client credentials.
- **To get started: Basic.** It works everywhere over TLS. SAP's design guidelines do not rate it the strongest choice.

With OAuth 2.0 on an on-premise system, the catalog flow needs two more services in the OAuth client's scope than the other flows: `/sap/opu/odata/IWFND/CATALOGSERVICE;v=2` and `/sap/opu/odata4/iwfnd/config/default/iwfnd/catalog/0002`. Without them the catalog flow answers `502 UPSTREAM_ERROR` while the connector still reads data.

Behind an SAP Cloud Connector

Set `S4_PROXY_TYPE` to `sapcc`, `S4_HOSTNAME` to `http://` plus the virtual host and port (for example `http://s4hana.virtual:44300`), and `S4_CC_LOCATION_ID` if you run more than one Cloud Connector. SAP requires `http://` on this route. Nothing travels unencrypted: the tenant reaches the Cloud Connector through its encrypted tunnel, and you choose the protocol of the last hop in the Cloud Connector.

All three flows were tested through SAP Cloud Connector 2.17.2 and behave as they do over the internet. In the Cloud Connector, under *Cloud To On-Premise* › *Access Control*:

Setting	Value
Back-end Type	ABAP System
Protocol	HTTPS if your SAP port is HTTPS, otherwise HTTP
Internal Host and Port	Your SAP system, by the name its certificate is issued to
Virtual Host and Port	A name of your choice. This is what goes into <code>S4_HOSTNAME</code>
Principal Type	None
Resource	<code>/sap/opu/odata</code> , <i>Path And All Sub-Paths</i> . It covers <code>/sap/opu/odata4</code> too

To test the route without any flow, use **Monitor › Manage Security › Connectivity Tests › Cloud Connector**. If it does not connect:

What you see	Cause
The mapping says <i>Not Reachable</i> at once	The protocol does not match the port
The mapping says <i>Not Reachable</i> after a few seconds, and <i>Configuration › On Premise › Backend Trust Store</i> shows no trusted backends	Add the certificate of the authority that signed your SAP system's certificate
The Cloud Connector says <i>Connected</i> , but every call answers 503 Timed out waiting for tunnel to open after about ten seconds	The Cloud Connector's Java runtime does not know the authority that signs the tunnel host of newer landscapes (<i>DigiCert TLS RSA4096 Root G5</i>). Its log <code>scc_core.trc</code> says <code>unable to find valid certification path</code> . Update that Java, or add the root to its trust list, and restart the Cloud Connector
503 There is no SAP Cloud Connector (SCC) connected ..., within a second	No Cloud Connector is paired under that Location ID
403 Access denied to resource ...	The path is not among the mapping's resources
405 HTTPS proxying is not supported	S4_HOSTNAME starts with <code>https://</code> . On this route it is <code>http://</code>

The 403, 405 and 503 answers come from SAP's platform, and the connector passes them on as they are.

S/4HANA Cloud: which scenario grants which API

Create one communication arrangement per scenario. The common ones:

Data	OData service	Communication scenario
Business partners, customers and suppliers	API_BUSINESS_PARTNER	SAP_COM_0008 Business Partner, Customer and Supplier Integration
Products	API_PRODUCT_SRV	SAP_COM_0009 Product Integration
Sales orders	API_SALES_ORDER_SRV	SAP_COM_0109 Sales Order Integration
Purchase orders	API_PURCHASEORDER_PROCESS_SRV	SAP_COM_0053 Purchase Order Integration
Supplier invoices	API_SUPPLIERINVOICE_PROCESS_SRV	SAP_COM_0057 Supplier Invoice Integration
G/L accounts	API_GLACCOUNTINCHARTOFACCOUNTS_SRV	SAP_COM_0241 Finance - Cloud Consolidation Data Collection Integration, or SAP_COM_0087 SAP Analytics Cloud for Planning Integration
Journal entry items	API_JOURNALENTYITEMBASIC_SRV	SAP_COM_0241 or SAP_COM_0087, as above
G/L account line items	API_GLACCOUNTLINEITEM	SAP_COM_0303 Finance - Accounting Analytics Integration
The service list of the catalog flow	The communication arrangement API	SAP_COM_0A07

Customers and suppliers are read through the business partner service. `SAP_COM_0002`, Finance - Posting Integration, is for posting journal entries and does not grant reading them.

Source: the communication arrangements of an SAP S/4HANA Cloud Public Edition tenant of release 2608, read in September 2026 and checked against SAP Help (*Communication Scenarios, APIs for General Ledger Accounting*). For your release, the app *Display Communication Scenarios* in your own system is the authority.

3. The service list: the catalog flow

The catalog flow answers on `https://<runtime host>/http/keboola/catalog` with the OData services of your SAP system, each with its path. Keboola calls it when somebody clicks *Load Services*. Without this flow you type the service path by hand (§5).

`CATALOG_SOURCE` sets where the list comes from:

Value	Where the list comes from	Right for
auto (shipped)	The arrangements first, and the Gateway catalogs when the system has no arrangement API	Almost everyone
arrangements	The communication arrangements that name your SAP user. Needs an arrangement for <code>SAP_COM_0A07</code>	S/4HANA Cloud
gateway	SAP's Gateway catalogs, OData V2 and V4 together	On-premise and private edition

The response header `X-Keboola-Catalog-Source` says which one answered.

What to expect:

- **One call is one run** in **Monitor** → **Message Processing Logs**, and nothing is cached. With the shipped `HTTP_TIMEOUT_MS` of 10 seconds a call finishes within about 50 seconds. Keboola waits 60, so do not raise the timeout on this flow.
- **V2 and V4 services are listed together.** Each description ends in `(OData V2)` or `(OData V4)`. For a support call, `?odata=v2` or `?odata=v4` on the address keeps one kind.
- **On S/4HANA Cloud, do not set gateway.** The Gateway catalogs answer `403` there, and no arrangement changes that.
- **On S/4HANA Cloud the list is filtered by your SAP user.** With a client certificate, or when no arrangement names the user, the services of every arrangement are listed.
- **An on-premise list shows what exists, not what your user may read.** A listed service can still answer `403` when it is read. An OData V4 service can answer `500 Service is not registered` until it is activated on the Gateway (transaction `/IWFND/V4_ADMIN`).

Views (on-premise)

An on-premise Gateway lists everything, typically ten thousand entries. The flow answers with one of three views:

View	What it keeps	Typical size
interfaces (shipped)	Services SAP classifies as WEB_API, services you built yourself, anything named API_*, and released OData V4 services	a few hundred
extended	Everything SAP does not classify as a UI service	under a thousand
all	Everything the catalogs list	everything

Set the view in `CATALOG_VIEW`, or for one call with `?include=` on the address. The response header `X-Keboola-Catalog-View` says which view answered and how much it hid. On S/4HANA Cloud the three views are identical.

The address book

On S/4HANA Cloud an arrangement names an OData V4 service (`API_PRODUCT`) but not its path (`/sap/opu/odata4/sap/api_product/srvd_a2x/sap/product/0001`). The address book supplies the path and a title. It holds the 584 OData services SAP has released for S/4HANA Cloud as of September 2026. A service with several versions is listed once per version: the lowest as `API_PRODUCT`, the next as `API_PRODUCT_0002`.

`CATALOG_RESOLVE_V4` decides what happens to a V4 service of an arrangement:

Value	What happens
lookup+verify (shipped)	The address book first. A name it does not know is tried at up to four paths built from the name, each with one small read of the service document, never of data. At most <code>CATALOG_VERIFY_LIMIT</code> tries per call (shipped 20, allowed 0 to 200)
lookup	The address book only. A name it does not know is left out
off	No OData V4 service of an arrangement is listed

A service that is left out is always named: the response header `X-Keboola-Catalog-Unlisted` counts them, and the run lists them in the field `CatalogUnlistedNames`. The remedy is a row in the address book. SAP releases new services with every release, so expect to add one now and then.

You find the path of a service in the *Communication Arrangements* app of your S/4HANA Cloud system: open the arrangement, and under *Inbound Services* each service shows its URL. The part after the host is the path. A V2 service at `/sap/opu/odata/sap/<name>` needs no row.

To add a service (a new V4 service, a service of your own, or a better title):

1. **Design** → **the package** → **SAP S4HANA Service Addresses for Keboola** → **Edit**. The agency is Keboola on both sides.
2. Add a pair `SERVICE_NAME` → `SERVICE_PATH`: the name as the arrangement shows it, in upper case and without the suffix `_G4BA` (`API_PRODUCT`), and the path without the host.
3. Add a second pair `SERVICE_NAME` → `SERVICE_TITLE` with the same name and the title you want to see in Keboola.
4. **Save and deploy the value mapping**. The running catalog flow reads the new row within seconds. It is not redeployed.

Every title and every path must be unique. The runtime refuses to deploy a mapping in which two names share one, and says only that the deploy failed.

Keep a note of the rows you add: a later version of the package brings a new address book, and your rows have to be added again.

When the list cannot be read

A list that could not be read is an error. An empty list means SAP answered and listed nothing.

Answer	When
502 UPSTREAM_ERROR	SAP refused the sign-in, or answered with a server error. On S/4HANA Cloud: the user has no arrangement for SAP_COM_0A07
500 CONFIG_ERROR	S4_HOSTNAME is not the API address of the system: it names a sign-in page, a web front end or a redirect. The error text says what to enter
502 UPSTREAM_UNREACHABLE	No such host, connection refused, or the Cloud Connector is not connected
504 UPSTREAM_TIMEOUT	A call took longer than HTTP_TIMEOUT_MS, or reading the arrangements took over 35 seconds
200 with the custom status CATALOG_EMPTY	SAP answered and listed nothing. Almost always an SAP user not yet assigned to an arrangement

For a support call, set `CATALOG_DIAGNOSTICS` to `true` and add `?debug=1` to the address: the answer then lists every read of SAP with its status and duration, and the services left out. Leave the parameter at `false` otherwise, because every caller with the role could read the diagnostics. The run always carries the same notes as the log property `CatalogDiagnostics`.

4. Reading data: the connector

Keboola calls `https://<runtime host>/http/keboola/connector/<the full OData path>`. The part after `/connector/` goes to SAP unchanged, so it must be the complete OData path, for example `/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_BusinessPartner`.

How it behaves:

- **It only reads.** It answers `GET` and `HEAD`. Reads need no CSRF token. Any other verb is refused, with `403` by the platform or `405` by the flow.
- **One request is one call to SAP.** SAP's status and body go back to Keboola untouched.
- **Only OData paths are forwarded.** `CONNECTOR_PATH_PREFIXES` ships as `/sap/opu/odata/`, `/sap/opu/odata4/`. A path outside them is answered `403 PATH_NOT_ALLOWED` and never reaches SAP. Add a prefix for a service that lives elsewhere. An empty value forwards every path, and then only the SAP user's authorizations limit what can be read.
- **Only the headers an OData client needs are passed on.** To SAP: `Accept`, `Accept-Language`, `Prefer`, `If-None-Match` and `If-Modified-Since`. Back to Keboola: the content, caching and OData version headers, `sap-message`, `Location` and `Retry-After`.
- **Some paths never reach the flow.** The platform answers `400` to a path with an encoded slash (`%2F`), backslash (`%5C`) or percent sign (`%25`), a double slash or `/. ./`. An OData key that holds such a character cannot be read through the connector.
- **Addresses inside SAP's answers are not rewritten.** A next-page link names your SAP system, not the connector (§5).

What can be read is decided in four places: the path prefixes, the authorizations of the SAP user, the resources released in the Cloud Connector, and the role that guards the endpoint.

Errors and retries

The connector does not retry. Keboola does: its component retries `500`, `502`, `503` and `504` three times.

What happened	What Keboola receives	Does Keboola retry?
SAP answered	SAP's own status and body	5xx yes, 4xx no
SAP could not be reached	502 UPSTREAM_UNREACHABLE with the cause	Yes
SAP did not answer within HTTP_TIMEOUT_MS	504 UPSTREAM_TIMEOUT	Yes
The path is outside <code>CONNECTOR_PATH_PREFIXES</code>	403 PATH_NOT_ALLOWED	No. Add the prefix or correct the path
The flow is misconfigured, or the security material is missing or of the wrong type	500 CONFIG_ERROR naming the parameter	Yes, uselessly. Fix the setting
The platform failed inside the connector	500 CONNECTOR_ERROR naming the place	Yes. Send the run's <code>ErrorLocation</code> field to support

Every error body carries `messageId`, the message ID of the run in **Monitor** → **Message Processing Logs**. Keboola's job log does not show it (§5).

Keep `HTTP_TIMEOUT_MS` (shipped `50000`) below the component's timeout of 60 seconds. A slow SAP system then produces a `504` that Keboola can retry, not a dropped connection.

How large an answer can be

The largest answers tested are pages of 5000 wide records: 22.6 MB of JSON in 5.5 seconds, and 28.7 MB in 9.6 seconds. A page of 5000 narrow records is 2 MB in about a second. The connector holds one answer in memory and sets no limit of its own. The size of one answer is set by the page size Keboola asks for and by SAP's own page cap (§5).

5. Set up Keboola

In your Keboola project, add the component **SAP OData** (`keboola.ex-sap-odata`) and use these values.

Connection

Field	Value
Base URL	The url of the service key, with no path after it
Catalog Path	<code>http/keboola/catalog</code> . Leave empty if you do not deploy the catalog flow
Authentication Type	<i>Basic</i> , with the service key's <code>clientId</code> as the user name and its <code>clientsecret</code> as the password
Request Timeout	60 seconds, the default
Max Retries	3, the default. Every retry is a billed message (§8)
Default Page Size	1000, the default. See 1 below

For each table you read

Field	Value
Service Path	<code>http/keboola/connector/</code> plus the service path, for example <code>http/keboola/connector/sap/opu/odata/sap/API_BUSINESS_PARTNER</code> . <i>Load Services</i> fills it in
OData Version	v2 or v4, as the service list's description says. Never auto (2 below)
Entity set, fields, filter, load type, primary key	Your choice, for example the entity set <code>A_BusinessPartner</code>

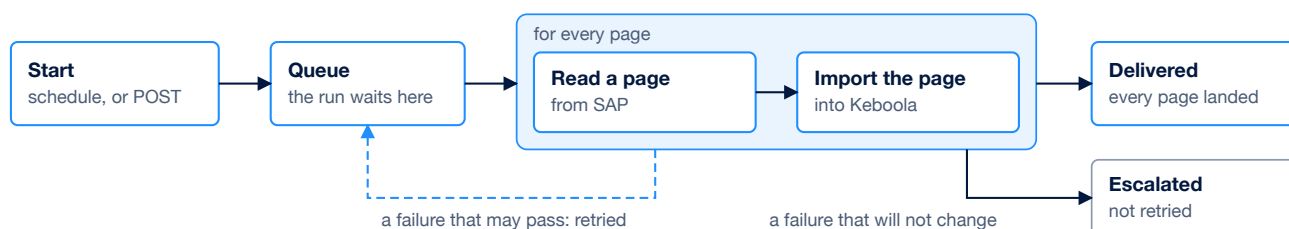
Four things to know about the component

1. **Keep the page size at or below your SAP Gateway's page cap.** A Gateway can cap a page at its own number of records, whatever is asked for, and send the rest as a next-page link that names your SAP system. The component follows the link and leaves the connector, and the job fails with 401 or an unknown host. On the on-premise test system the cap is 5000 records. To find yours, read one entity set with a large `$top` through the connector and look for `__next` in the answer.
2. **Set the OData version on every table.** On `auto`, a `$metadata` read that fails for a moment makes the component assume V2. A V4 service then delivers no rows and no error.
3. **Test Connection does not test a service path.** It calls the tenant and the catalog flow only. Use *Preview Data* or a small job to prove a path.
4. **The connector's error text is not in the job log.** The job shows the status and the address. The `code`, `message` and `messageId` are in the run in **Monitor** → **Message Processing Logs**. Find the run by time, or search by ID for the path that was read.

Load Services calls the catalog flow twice per click, so expect two runs. A *Load Services* failure that leaves no run never reached your tenant. Try again.

6. Scheduled delivery

The delivery flow reads one entity set from SAP and writes it into one Keboola table. A run starts on the schedule, or when you `POST` to the flow's endpoint. The run waits on a queue, then reads SAP page by page and imports each page into Keboola.



- **A failure that may pass** (a server error, a timeout, a dropped line) puts the run back on the queue. The platform retries it with growing waits.
- **A failure that will not change** (not authorized, a wrong path, a wrong table) ends the run with the status **Escalated**. It is not retried. Fix the cause, and the next scheduled run reads the same data again.

Settings

The shipped values read business partners. `SERVICE_PATH`, `ENTITY_SET`, `DELTA_FIELD` and `PRIMARY_KEY` are one example that works together. To read another entity set, change all four.

What to read

Parameter	Shipped	Meaning
SERVICE_PATH	/sap/opu/odata/sap/API_BUSINESS_PARTNER	Path of the service, without the host
ENTITY_SET	A_BusinessPartner	The entity set to read
ODATA_SELECT	empty	Fields to read, separated by commas. Empty reads every field. Must include the columns of PRIMARY_KEY
ODATA_FILTER	empty	An extra condition, in OData's filter language
PAGE_SIZE	5000	Records per request, 1 to 20000

How to load

Parameter	Shipped	Meaning
LOAD_MODE	full	full replaces the table on every run. incremental updates it by primary key
PRIMARY_KEY	BusinessPartner	Columns that identify one record, separated by commas. Required for incremental loading
DELTA_FIELD	LastChangeDate, CreationDate	Fields that say when a record changed, separated by commas
DELTA_FIELD_TYPE	datetime	How those fields are typed: datetime, datetimeoffset or date
DELTA_PRECISION	day	day if SAP records only the date of a change, second if it records the time in universal time
DELTA_OVERLAP_MINUTES	15	At second precision: how far before the previous run's start the next run begins reading. 0 to 10080

Where to write

Parameter	Shipped	Meaning
KEB00LA_STACK_URL	empty	The address of your Keboola stack, for example https://connection.eu-central-1.keboola.com . Must start with https://
KEB00LA_TABLE_ID	empty	The table as Keboola shows it, for example <code>in.c-sap.business_partner</code> . The bucket must exist. The flow creates the table
KEB00LA_CREDENTIAL_ALIAS	KEB00LA_STORAGE	Security material with the Storage token
KEB00LA_IMPORTER_URL	empty	Only to override the import address looked up from the stack

When and how it runs

Parameter	Shipped	Meaning
DELIVERY_SCHEDULE	every day at 02:00 UTC	When a run starts on its own. Set with the scheduler control on the Configure screen
DELIVERY_ADDRESS	/keboola/deliver	The flow's own endpoint: <code>https://<runtime host>/http/keboola/deliver</code>
DELIVERY_QUEUE_NAME	KeboolaDelivery	The queue that holds the runs
DELIVERY_RETRY_INTERVAL_MIN	3	Minutes to wait before the first retry. Doubles each time
DELIVERY_MAX_RETRY_INTERVAL_MIN	60	Longest wait between retries, in minutes
HTTP_TIMEOUT_MS	120000	Longest one call may take, to SAP or to Keboola. 1000 to 3600000

Starting a run

- **On the schedule**, at the time in `DELIVERY_SCHEDULE`. Deploying the flow starts no run.
- **By hand**. POST to `https://<runtime host>/http/keboola/deliver`, signed in with the service key. It answers 202 and the `messageId` of the run it queued. The endpoint asks for no CSRF token.

The settings are checked before anything is queued. A start with a missing or wrong setting is refused and names the setting: by hand with 500 `CONFIG_ERROR`, on the schedule as a run with the status *Failed*.

A queued run keeps the settings of *What to read* and *How to load*, and its table, as they were when it was queued. The connection settings are read again at each attempt.

Full or incremental

A full load replaces the table. The first page clears the table and every later page is appended. It is **not atomic**: for the minutes a run takes, a reader sees a partial table, and a full load that fails part-way leaves the table partial until a retry or the next run completes. If something reads the table continuously, use incremental loading. A full load that reads no records leaves the table as it was.

An incremental load reads the period since the watermark, the start of the previous successful run, and updates the table by primary key.

- **Name the creation field too.** Many SAP services leave the change field empty until a record is first changed. `DELTA_FIELD` takes several fields, and a record is read when any of them falls in the period.
- **The period overlaps on purpose.** At day precision it opens one day early, because SAP's dates are in the SAP system's time zone. At second precision it opens `DELTA_OVERLAP_MINUTES` early, because SAP can save a change slightly later than its time stamp.
- **Deletions are not carried over.** A record deleted in SAP stays in the Keboola table. Run a full load to drop it.
- **The watermark is saved only after every page has landed**, so a run that fails half way reads its whole period again. The watermark lives in the data store `KeboolaDeliveryState` (**Monitor** → **Manage Stores** → **Data Stores**). Delete the entry, and the next run reads everything.

Set a primary key

With `PRIMARY_KEY` set, a record delivered twice updates itself, and the pages are read in key order. Without one, a redelivered record is added again, and the run is marked `PagingUnsorted = true`. Set the key whenever you can, and always for incremental loading.

- The key is given to the table when the flow creates it. If the table exists with another key, the run ends `KEB00LA_FAILED` and names both keys.
- Some services have a key SAP cannot sort by (the journal entry items, for one). The run then ends Escalated with SAP's error text. Read such a service without a key.

The table

If the table does not exist, the flow creates it from the columns of the first page, with `PRIMARY_KEY` as its key. **It is a typed table and every column is `STRING`**. Convert in Keboola: a wrong type is harder to undo than a conversion. Column names are limited to 64 characters by Keboola. A longer one ends the run before anything is imported, and names the column.

OData V2 services return dates as `/Date(1777939200000)/` and times as `PT19H02M07S`. The delivery flow writes dates as `yyyy-MM-ddTHH:mm:ss` (no time zone, as SAP's value has none) and times as `19:02:07`. Numbers are written in plain notation.

More than one table

One flow delivers one entity set into one table. For another table, copy the flow within the package and give the copy three things of its own on its Configure screen:

1. **An endpoint** in `DELIVERY_ADDRESS`, for example `/keboola/deliver/products`.
2. **A queue** in `DELIVERY_QUEUE_NAME`. Copies that share a queue carry out each other's runs. Each queue counts against the tenant's queue quota. **Monitor** → **Manage Message Queues** shows what is in use.
3. **A table** in `KEB00LA_TABLE_ID`. Two deliveries must never write into the same table.

Sizes and run times

Tested on an on-premise system with full loads of 190 134 journal entry items of 149 columns, about 5.7 KB of JSON a row, the widest record on that system:

PAGE_SIZE	Pages	Run time	Rows per second	Size of one page
1000	191	65 minutes	49	5.7 MB
5000	39	17 minutes	184	29 MB
20000	10	8 minutes	415	115 MB

- **The cost of a load is the number of pages.** Every page is one Keboola import of about ten seconds, plus the time SAP needs to answer.
- **Use 5000**, the shipped size, for rows of a few kilobytes. **20000 is the ceiling:** use it for narrow rows. A wide record at 20000 means pages over 100 MB in memory, which leaves little room for a second large flow on the tenant. Use 1000 only when the record is very wide or SAP is slow per row.
- **One run must finish within four hours**, the time the queue keeps a run locked. A run that outlives the lock can be handed out a second time. At 5000 rows a page that is about 550 pages, or 2.7 million rows of this width. Narrower rows go faster.
- **Split larger loads.** Read them through the connector, or over copies of the flow, each with its own `ODATA_FILTER`.
- **One run per table at a time.** Start a second run when the first shows `DELIVERED`.

What is retried

What happened	Status of the run	Retried?
SAP or Keboola answered with a server error, 408 or 429, did not answer in time, or the line dropped	<i>Retry</i>	Yes, by the platform, until it succeeds or the queued run expires after 90 days
The security material for SAP is missing or of the wrong type	<i>Retry</i>	Yes, until you correct the material
SAP answered 4xx (403 no arrangement, 404 wrong path, 400 bad filter), or with something that is not a list of records	<i>Escalated</i> , <code>UPSTREAM_FAILED</code>	No. Fix the cause
SAP returned the same records twice in a row: the service ignores <code>\$skip</code> , or <code>PRIMARY_KEY</code> does not identify one record	<i>Escalated</i> , <code>UPSTREAM_FAILED</code>	No. Check <code>PRIMARY_KEY</code> . If it is right, set <code>PAGE_SIZE</code> above the number of records
Keboola refused the token, the table or the import, or answered with a redirect	<i>Escalated</i> , <code>KEB00LA_FAILED</code>	No. The Storage token is never sent after a redirect
The table's key differs from <code>PRIMARY_KEY</code> , or a column name is over 64 characters	<i>Escalated</i> , <code>KEB00LA_FAILED</code>	No. Nothing is imported
The run needs more than 99 999 pages	<i>Escalated</i> , <code>CONFIG_ERROR</code>	No. Raise <code>PAGE_SIZE</code> or narrow <code>ODATA_FILTER</code>
The rows landed, but the watermark could not be saved	<i>Completed</i> , <code>DELIVERED_WATERMARK_FAILED</code>	Not needed. The next run reads the period again, and the primary key turns repeats into updates

A run that waits for a retry is listed in **Monitor** → **Manage Message Queues** with its retry count and next retry time. Retry or delete it there once the cause is fixed.

A waiting run keeps the `ODATA_SELECT`, `ODATA_FILTER` and `PAGE_SIZE` it was queued with. When SAP cannot serve a page at all (a 500 for certain records), the run is retried again and again. Delete the waiting run, narrow `ODATA_SELECT` or `ODATA_FILTER`, and start a new run.

Limits to know

- **A retried run skips the pages that already landed.** One case escapes: a page Keboola finished importing after the flow stopped waiting is imported again. With a primary key its rows replace themselves. Without one they are in the table twice until the next full load.
- **Paging is by offset.** Rows inserted in SAP between two attempts of the same run can shift the page boundaries, so a skipped page may not hold exactly the rows it held the first time. The next incremental run reads them.
- **The end of the data is SAP's word.** An answer with fewer records than asked for and no next-page link is taken as the end. A service that cuts its answers short without a link loses the rest unnoticed.
- **A watermark that cannot be read is not silent.** The run reads everything and is marked `WatermarkRead = failed, full window`.

7. Monitoring and troubleshooting

Every run is in **Monitor** → **Message Processing Logs**. A delivery has two entries: the start (QUEUED) and the run itself. Two columns matter: the platform's **status** and the package's **custom status**.

Custom status	Meaning	What to do
OK, QUEUED, DELIVERED	A read was answered, a delivery was queued, a delivery landed	Nothing
UPSTREAM_FAILED	SAP refused the request and said why	Read the fields <code>UpstreamStatus</code> and <code>UpstreamError</code> . Usually a missing communication arrangement
UPSTREAM_UNREACHABLE, UPSTREAM_TIMEOUT	SAP could not be reached, or did not answer in time	Check <code>S4_HOSTNAME</code> , the Cloud Connector and the network. For a timeout, SAP's health and <code>HTTP_TIMEOUT_MS</code>
UPSTREAM_ERROR	The catalog flow could not read its list	The error text says why: a refused sign-in, the OAuth scope (§2), or <code>SAP_COM_0A07</code> (§3)
KEB00LA_FAILED	Keboola refused the import or the table	Read <code>KeboolaStatus</code> , <code>KeboolaError</code> and the attachment <code>ErrorDetails</code> : token, bucket rights, table, primary key, column name
CONFIG_ERROR	The flow is misconfigured	The error text names the parameter
PATH_NOT_ALLOWED	The connector was asked for a path outside its prefixes	Correct the service path in Keboola, or add the prefix (§4)
CATALOG_EMPTY	The catalog flow found nothing	Assign the SAP user to an arrangement
DELIVERED_WATERMARK_FAILED	Every page landed, but the watermark could not be saved	Read the attachment <code>WatermarkError</code> . The data is in Keboola

Custom status	Meaning	What to do
CONNECTOR_ERROR, CATALOG_ERROR	The platform itself failed inside a flow	Send the error text, the field <code>ErrorLocation</code> and the attachment <code>ErrorDetails</code> to support
METHOD_NOT_ALLOWED, DELIVERY_NOT_STARTED	The caller's request was refused	The error text says why

For three platform statuses of the delivery flow the package sets no custom status. Look for them in the Status column:

Status	Meaning	What to do
<i>Retry</i>	A run failed for a reason that may pass and waits on the queue	Nothing, unless it lasts. See Manage Message Queues
<i>Failed</i>	A scheduled delivery was refused at its start	The error text names the setting (§6)
<i>Discarded</i>	Beside every scheduled run: the same start on the tenant's second node, suppressed	Nothing. It is not a failure

A delivery run that ended with a failure that will not change shows *Escalated* and its custom status side by side. The reason is in the attachment `ErrorDetails` and in the field `KeboolaError` or `UpstreamError`.

Alerting

The package sends no alerts. Your tenant's monitoring does. Watch for:

- **Status Escalated:** a delivery run whose data did not arrive. In the tenant's API: `MessageProcessingLogs? $filter=Status eq 'ESCALATED'`. Add and `CustomStatus eq 'KEB00LA_FAILED'` to watch one cause.
- **Status Retry that lasts** longer than a few retry intervals.
- **The custom status `DELIVERED_WATERMARK_FAILED`** on a completed run.
- **For pull, use Keboola's own job notifications.** Every refused read fails the Keboola job in the same second.

SAP's tool for this is **SAP Cloud ALM** (*Integration and Exception Monitoring*). It reads the tenant's message logs and alerts on failed and escalated messages by e-mail, chat or ticket. A lighter way is a small flow of your own on a timer that reads the query above and posts to the **SAP Alert Notification service**.

Where a 401 or 403 comes from

A 401 or 403 in a Keboola job comes from one of these:

The run in the monitor	Who refused
Exists, with UPSTREAM_FAILED, and the field UpstreamError begins Access denied to resource	The Cloud Connector's mapping (§2)
Exists, with UPSTREAM_FAILED, and UpstreamError is SAP's own error text	SAP refused the user of the security material
Exists, with PATH_NOT_ALLOWED	The service path is outside the connector's prefixes
Does not exist	Your tenant refused Keboola's key or role (§1), or the component followed a next-page link past the connector (§5)

What a run records

These are searchable fields, so you can filter on them and read them by API. The platform keeps the first 200 characters of each.

Flow	Fields
Connector	HttpPath, UpstreamStatus, UpstreamOutcome, UpstreamErrorCode, UpstreamError, ErrorLocation
Catalog flow	CatalogSource, CatalogCount, CatalogHidden, CatalogView, CatalogUnlisted, CatalogUnlistedNames
Delivery flow	Pages, RowsDelivered, TableId, Watermark, DeltaWindow, DeltaOverlap, FirstPageQuery, WatermarkRead, PagingUnsorted, UpstreamStatus, UpstreamError, KeboolaStatus, KeboolaError

SAP's answer to a read goes back to Keboola and is not kept in the log. Credentials are redacted wherever a request is logged.

8. Good to know

Scope. The package is built and tested for SAP S/4HANA, Cloud and on-premise. Other SAP products, SAP IBP among them, are not covered by this version.

Retries are billed. One execution of a flow counts as one message, however many calls to SAP it makes (SAP Note 2942344). Every retry is another message, which is why a failure that will not change is not retried. The queue itself is not metered. The same note announces metering by size, which was not switched on at the time of writing.

Data residency. The data moves from your SAP system to your Integration Suite tenant to your Keboola project. Those three can be in different regions. Choose them deliberately.

The integration key. AIR_KEY and AIR_HEADER_NAME identify this integration to SAP, once SAP has issued a key for it. Leave both empty unless Keboola gives you values. The key grants nothing and is not a secret.

9. Rotating credentials

No credential needs the flow editor.

Credential	Where it lives	How to rotate
The SAP user's password or client secret	The security material in S4_CREDENTIAL_ALIAS	Set the new secret in SAP. Under Monitor → Manage Security → Security Material , edit the entry, enter the new value and deploy it. Redeploy every flow that uses it. Do all of it in one sitting: in between, reads answer 401 and a delivery run ends Escalated
The client certificate	The key pair in S4_PRIVATE_KEY_ALIAS	Add the renewed key pair to the keystore, map its certificate to the SAP user, and redeploy the flows
The Keboola Storage token	The security material in KEB00LA_CREDENTIAL_ALIAS	Create the new token in Keboola, edit the material's password, deploy it, redeploy the delivery flow. Only then revoke the old token
The service key Keboola signs in with	The service instance in the BTP cockpit, and the component in Keboola	Create a second key on the same instance, enter it in the component, run one job, then delete the old key. No flow changes

10. The Keboola adapter

The adapter is for integration flows you build yourself. Your flow hands it a message body of comma separated values with a header line, and the adapter writes those rows into one Keboola Storage table. It reads nothing from SAP, and it does not page, queue or retry. To get SAP data into Keboola without building anything, use the delivery flow, which does not need the adapter.

Deploy the adapter before the flow that uses it. It needs *User Credentials* with the Storage token as the password, as in §1.

Setting	Shipped	Meaning
Keboola Stack URL	empty	The address of your Keboola stack. Must start with <code>https://</code>
Credential Name	KEB00LA_STORAGE	The name of the User Credentials
Table	empty	The full table identifier. The table must exist. The adapter does not create it
Incremental Load	on	On: the rows are added. Off: the table's content is replaced
Timeout (in ms)	60000	How long to wait for Keboola, which answers when the import is finished

On success the message body is Keboola's answer, and the adapter sets the headers `KeboolaHttpStatus`, `KeboolaTableId` and `KeboolaIncremental`. A failed delivery raises an exception in your flow, and its error text names the cause:

Error text	What to do
... failed with status 401: Keboola did not accept the Storage token	The token is wrong, expired or revoked
... failed with status 404: Table ... not found	Create the table, or correct <i>Table</i>
Could not find credential for the given alias ...	<i>Credential Name</i> names nothing that is deployed
Keboola Stack URL must start with https://...	Correct the address
Could not reach the Keboola importer at ...: Read timed out	The import may still have happened. Check the table before sending the same rows again
... redirects are not followed	Something answers in Keboola's place. The token is sent to the configured stack only

The adapter sends one message as one file and holds it in memory. Keep a delivery to tens of megabytes and split larger data into several messages.

11. Support and responsibilities

Keboola publishes and supports this package. Report a problem to Keboola's support channel, as named on the package's listing on the SAP Business Accelerator Hub. Send with it:

- the flow and its version,
- the custom status and the `messageId` of the run,
- the run's fields and its `ErrorDetails` attachment,
- what was configured, but never the security material,
- for a Keboola job, the job id.

Who	Is responsible for
Keboola	The package as delivered, this guide, new versions on the Hub, and the SAP OData component in Keboola
SAP	The Integration Suite platform, the Cloud Connector, and SAP S/4HANA and its APIs
You	Your tenant and its sizing, the security material and keys, the SAP user and its arrangements, the Cloud Connector's mappings, the Configure values, the rows you add to the address book, and your Keboola project

A flow you have edited in the flow editor no longer receives the package's updates from the Hub, and support starts from the unedited version.

12. Change log

Version 1.1.0

All flows

- The sign-in method to SAP is a Configure value, `S4_AUTH_METHOD`. One build serves OAuth 2.0 client credentials, Basic and a client certificate. The first version required an edit of each flow.
- The role a caller must hold is a Configure value, `SENDER_ROLE`.
- Only the request headers an OData client needs are passed on to SAP.
- A failure of the platform is reported as `CONNECTOR_ERROR` or `CATALOG_ERROR`, no longer as a sign-in failure.

The connector

- `CONNECTOR_PATH_PREFIXES` limits what is forwarded to the OData paths.
- The tested maximum answer is documented.

The catalog flow

- Rebuilt. It reads SAP through the flow's own connections, so the proxy type, the Cloud Connector and the sign-in method of the Configure screen apply.
- It no longer probes services with data reads, and it caches nothing.
- The address book is a value mapping you can maintain. It replaces the built-in service list.
- A wrong host is a `500 CONFIG_ERROR`, never an empty list.
- As shipped, one call finishes within about 50 seconds.
- The switches `?mode`, `?nocache` and `?filter` are gone. `?debug=1` answers only with `CATALOG_DIAGNOSTICS=true`.

The delivery flow

- A failure that will not change ends the run Escalated, so the monitor shows it.
- `DELIVERY_ADDRESS` and `DELIVERY_SCHEDULE` are Configure values. A deploy starts no run.
- Incremental loading at second precision overlaps the previous period by `DELTA_OVERLAP_MINUTES`. A watermark that cannot be read or saved is named in the monitor.
- `PAGE_SIZE` ships as 5000, with a ceiling of 20000. Tested sizes and run times are documented.
- Dates are written without a time zone suffix. A column name over 64 characters ends the run before any import.
- The way rows are written to Keboola is unchanged: one import per page. The adapter is unchanged too.

This guide

- New: setting up Keboola, alerting, rotating credentials, support and responsibilities, this change log.

Version 1.0.0

- First release.

Appendix A. Signing in to SAP S/4HANA, step by step

This appendix covers the sign-in from the flows to SAP. Which method suits which system is in §2. How Keboola signs in to your tenant is in §1.

OAuth 2.0 Client Credentials (shipped)

1. **Monitor** → **Manage Security** → **Security Material** → **Create** → **OAuth2 Client Credentials**.
2. Fill in:

Field	Value
Name	KEB00LA_S4
Token Service URL	The token address of your SAP system. On-premise and private edition: <code>https://<host>/sap/bc/sec/oauth2/token?sap-client=<client></code> . S/4HANA Cloud: the token endpoint the communication arrangement shows
Client ID, Client Secret	The OAuth client registered on the SAP system, or the client id and secret the arrangement issued
Client Authentication	Send as request header, unless your SAP system expects it in the body

3. Deploy the security material.
4. Leave `S4_AUTH_METHOD` and `S4_CREDENTIAL_ALIAS` on the Configure screen as shipped.

Basic

1. **Monitor** → **Manage Security** → **Security Material** → **Create** → **User Credentials**. Name it `KEB00LA_S4`, and enter the SAP user and its password.
2. Deploy the security material.
3. On the Configure screen of each flow, set `S4_AUTH_METHOD` to *Basic*.

Client Certificate

1. **Monitor** → **Manage Security** → **Keystore**. Add a key pair, or create one, and note its alias.
2. On the SAP system, assign the public certificate of that key pair to your SAP user.
3. On the Configure screen of each flow, set `S4_AUTH_METHOD` to *Client Certificate* and `S4_PRIVATE_KEY_ALIAS` to the alias of the key pair. `S4_CREDENTIAL_ALIAS` is not used.

This method cannot be used behind an SAP Cloud Connector (§2).

When the sign-in does not work

What you see	Cause
500 CONFIG_ERROR naming S4_CREDENTIAL_ALIAS and S4_AUTH_METHOD, from the connector or the catalog flow	The security material is missing, or of another type than the method needs
A delivery run that stays in the status <i>Retry</i>	The same fault in the delivery flow. The run completes at its next retry once the material is corrected
401 from SAP with UPSTREAM_FAILED on the connector, or 502 UPSTREAM_ERROR on the catalog flow	SAP refused the user, the password or the token, or the user is locked in SAP
502 UPSTREAM_ERROR from the catalog flow naming a 403 and a 500, while the connector reads	OAuth 2.0 against an on-premise system whose OAuth client lacks the scope of the two Gateway catalog services (§2)
The deploy of a flow fails after you changed the method	S4_AUTH_METHOD holds a value that is not in the list. The copy that was running before keeps answering until you correct it

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